

One Game, One Experience: Shared-Control Mechanisms for Cross-Platform Multiplayer Games

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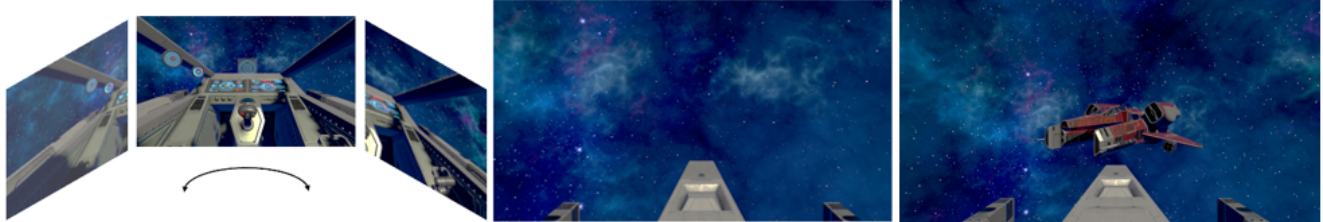


Figure 1: Our shared-control game, *Spaceship Fighter*. The VR player, positioned inside the spaceship, pilots and navigates it (left), while the non-VR player, viewing the spaceship from above (middle), is responsible for shooting enemies (right).

Abstract

Cross-platform games prominently involve multiplayer interactions between VR and non-VR players, presenting unique challenges in balancing player experiences (PX). Many game designs often prioritise the VR player's immersive role, while assigning non-VR players supportive tasks. Consequently, non-VR players often experience a less immersive and sometimes incomplete game experience. To examine whether shared-control mechanisms could address these discrepancies, we designed *Spaceship Fighter* as a novel approach to balancing PX in cross-platform games. Ten participant pairs played the game ($n=20$), with each participant taking turns as both the VR and non-VR player, to collaboratively control different functions of a single in-game entity. We found that the majority of players perceived both roles as equally important and enjoyable, suggesting that the shared-control mechanism successfully balanced role significance. These findings demonstrate the potential of shared-control mechanisms to enhance inclusivity and collaboration in cross-platform multiplayer games, offering a novel solution to promote balanced player experiences.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **Empirical studies in HCI**; **Collaborative interaction**; • **Software and its engineering** → **Interactive games**.

Keywords

Virtual Reality, Video Games, Multiplayer, Cross-platform, Shared-Control

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1 Introduction

Cross-platform multiplayer games, which integrate VR and non-VR players, face significant challenges in balancing player experiences (PX). Existing research has investigated different techniques to balance PX in cross-platform play. The cooperative asymmetrical design method assigns distinct but interdependent roles to VR and non-VR players. For instance, VR players navigated a maze guided by non-VR participants [9], or puzzle-solving games required both roles to collaborate on shared goals [2, 10, 13]. However, non-VR participants often feel peripheral and less engaged compared to the immersive roles assigned to VR players [12]. Other research has focused on integrating external hardware and interaction techniques to enhance non-VR player engagement. For example, Gugenheimer et al. [6] introduced mobile displays and floor projections to connect non-VR players to VR environments, enabling them to visualise and interact with the virtual world. While this approach improved engagement, non-VR players remained in supportive roles with limited immersion. In another approach, touch-sensitive displays on VR headsets allowed non-VR users to interact through gestures, such as manipulating virtual objects, which enhanced their involvement but introduced challenges like potential safety risks [7].

A shared-control mechanism has shown promise as a solution to cross-platform play by enabling players to collaboratively control a

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single entity or system [15]. Despite its success in some multiplayer games on gaming consoles or desktops, shared-control has not been explored in VR and non-VR cross-platform contexts. This gave rise to our research questions: **to what extent can a shared-control mechanism enable balanced engagement and equitable participation between VR and non-VR players in cross-platform games? How do players perceive their roles and interactions when engaging in a shared-control mechanism across different platforms?**

This paper investigates whether a shared-control mechanism can enable balanced engagement and equitable participation across platforms, while exploring players' perceptions of their roles in such setups. We designed a novel shared-control mechanism cross-platform game called *Spaceship Fighter* (Figure 1). We employed a user study with 20 participants to assess the mechanism's effectiveness in scaffolding shared control. We found that shared-control addresses disparities in PX by creating interdependent and intuitive interactions for both VR and non-VR players. Participants reported PX and preferences across platforms, indicating balanced gameplay. The findings demonstrate the potential of this novel approach for addressing the PX gap between VR and non-VR platforms.

2 Related Work

Our research draws on three key areas of exploration: (1) challenges in balancing player experience (PX) in cross-platform VR and non-VR gameplay, (2) methods addressing these challenges, and (3) shared-control mechanisms in game design. We synthesised related work across these areas to identify opportunities for improving collaboration and engagement in cross-platform games.

2.1 Challenges in Cross-Platform Gameplay

Cross-platform gameplay fosters inclusivity by integrating players across VR and non-VR platforms, but differences in input methods and player perception create challenges with games affording equivalent experiences. In multiplayer combat games, non-VR players typically rely on keyboards or gamepads, while VR players, though benefiting from immersive controls and environments, often face less precise control. This disparity frequently leads to PX imbalances due to differences in game difficulty [3, 4]. In collaborative games, VR enhances presence through head-mounted displays (HMDs) [12], offering a deeply engaging experience. Non-VR players, in contrast, frequently occupy supportive roles in puzzle-solving or escape-room games, such as analysing information or providing instructions. These roles, though crucial for collaboration, are less immersive and action-oriented, discouraging non-VR participation over time.

2.2 Approaches to Addressing Cross-Platform Gameplay Challenges

Efforts to bridge the PX gap between VR and non-VR players fall into two main categories: cooperative asymmetric gameplay and external engagement enhancement methods.

2.2.1 Cooperative Asymmetric Gameplay. Asymmetric game design assigns distinct roles tailored to each platform's strengths. For

example, modular escape room games [5] allow VR players to navigate immersive spaces while non-VR players solve complementary puzzles by dividing cooperative tasks based on the strengths of each platform. Similarly, Zamojski et al. [16] developed a VR experience called *Ohm VR* which employs modular designs in which VR users manipulate virtual components, and non-VR users provide analytical guidance. *LabXscape* is another example where cross-reality collaboration is scaffolded by enabling VR players to explore 3D spaces, while AR users interact with mixed-reality elements, and PC players manage data [11]. Designs that assign interdependent tasks appear to encourage collaboration but they often reinforce primary (VR) and secondary (non-VR) roles which could diminish the non-VR players' sense of contribution and engagement.

2.2.2 External Engagement Enhancement. External methods seek to bridge the experiential gap by providing tools for non-VR players to influence the VR environment. These kinds of methods seek to make modifications to the physical environment of players. While this approach improves engagement for non-VR players, their interaction remains limited compared to the immersive experience of VR players. The game *FaceDisplay* incorporates touch-sensitive screens on HMDs and this approach enables gesture-based interactions for non-VR players [7]. However, this approach introduced safety concerns and negatively impacted the PX of the VR player by disrupting their sense of immersion. Another approach was to offer tactile feedback mechanisms, such as synchronised floor vibrations for both VR and non-VR players [8], which was shown to enhance the player experience of co-presence. Visual and physical design alterations, as in *BirdQuestVR*, align narrative and gameplay elements to engage non-VR participants [14]. Although these methods improve engagement, their reliance on external tools limits scalability and consistency across game genres.

2.3 Shared-Control Mechanisms in Multiplayer Games

Shared control enables multiple players to collaboratively manage a single game entity and this approach has been shown to promote teamwork and equitable contributions [15], while also being an effective strategy for balancing roles during play. A key concept of shared control is Locus of Manipulation (LoM), which refers to the in-game position where a player asserts control over the game world [15]. Shared control can be categorised based on how LoM is distributed among players, either through distinct LoM, where each player has independent control over separate elements, or mutual LoM, where players jointly manipulate the same entity. Zhou et al. [17] demonstrated how shared control in VR fosters co-presence and satisfaction, as players can collectively navigate and manipulate objects. Our work extends shared-control mechanisms to cross-platform settings. We integrate VR and non-VR players in a shared-control game. By distributing control responsibilities, we aim to reduce PX disparities to foster equitable collaboration.

3 Shared-Control Game Design and Implementation

We designed a game called *Spaceship Fighter* to explore cross-platform shared-control play. *Spaceship Fighter* is a multiplayer

game designed for VR and PC. It allows players using each modality to collaboratively control a spaceship and engage in combat. The goal of the game is to survive enemy attacks.

The gameplay revolves around two primary roles: piloting and attacking, requiring tight collaboration between players for success. As shown in Figure 2, the VR player assumes the role of the pilot, seated in the cockpit of the spaceship. Using hand controllers, they control realistic spaceship manoeuvres, including yaw, pitch, and roll adjustments. This setup provides a first-person perspective and enhances immersion, giving the VR player a sense of directly controlling the spaceship from within. Precise piloting is critical, as it determines the positioning of the spaceship to counter incoming threats effectively. Meanwhile, the non-VR player takes on the role of the gunner, who is responsible for targeting and eliminating enemies. Using a mouse, they aim within a predefined range and shoot to defend the spaceship. The gunner's success depends heavily on the VR player's ability to manoeuvre the ship effectively, highlighting the interdependence between the two roles.

The shared control mechanism in *Spaceship Fighter* follows the distinct LoM with coherent entity model as classified by Sykownik et al. In this system, each player independently controls a specific aspect of the same game entity, creating a tightly integrated gameplay experience. While the control points of the two players remain distinct, their actions are inherently interdependent. Unlike independent dual-player systems, where each player performs self-sufficient tasks, this design reinforces cooperation, ensuring that neither player can function optimally without the other.

The game was developed for the Oculus Quest 2 using Unity version 2021.3.20f1. The XR Interaction Toolkit and XR Plugin Management enabled smooth VR interactions and controls for the Oculus Quest 2. For real-time multiplayer capabilities, the Photon Unity Networking 2 (PUN2) system was integrated, ensuring efficient communication and synchronisation between VR and non-VR players.

4 User Study

The study followed a between-subjects design conducted in a controlled environment. Participants played the game in pairs, with one using an Oculus Quest 2 and the other using a desktop PC. Players were seated at separate desks in the same room to facilitate verbal communication. Each pair completed two 10-minute gameplay sessions, switching roles after the first session to experience both perspectives.

Each gameplay session was structured as follows:

- (1) **Orientation:** Participants were briefed on objectives and roles, with one player using the VR setup and the other a desktop PC.
- (2) **Familiarisation:** Participants were allowed to adjust to their equipment and practice using the controls.
- (3) **Gameplay Sessions:** Participants played the game for 10 minutes.
- (4) **Role Switching:** Participants swapped roles and repeated the orientation process.
- (5) **Post-Experiment Evaluation:** Participants completed a questionnaire and participated in a semi-structured interview.

4.1 Participant Recruitment

Twenty participants (10 pairs) were recruited via the online university participant recruitment system. Eligibility criteria included being 18 or older with prior VR or 3D gaming experience.

4.2 Data Collection and Analysis

Quantitative data were gathered using the *Player Experience Inventory* (PXI) [1], a 7-point Likert-style questionnaire. Custom questions assessed platform preferences and comfort levels. Statistical analyses with Mann-Whitney U tests, were used to identify significant differences in player experiences between VR and non-VR platforms.

Qualitative data were collected through semi-structured interviews, where participants provided feedback on collaboration, role clarity, and gameplay challenges. Responses were transcribed and coded to identify key themes, such as communication and engagement dynamics.

5 Results

5.1 Quantitative Data Analysis

A total of 20 completed questionnaires were collected from participants. The data from the 7-point Likert scale responses were encoded as numeric values from 1 (*Strongly Disagree*) to 7 (*Strongly Agree*). The Player Experience Inventory (PXI) questionnaire evaluated player experience across 10 criteria, with each criterion assessed through three mixed-order questions to encourage unbiased responses. Mean values for each criterion were calculated separately for VR and non-VR experiences. Statistical significance between the two settings was assessed using the Mann-Whitney U test, a non-parametric test suitable for comparing independent groups. Participants were also asked to state their platform preference on a five-point scale, ranging from *Much more VR* to *Much more Computer*.

Table 1 summarises the mean scores, differences, U-statistics, and p-values for all criteria, with bold font indicating statistical significant differences.

The *Immersion* criterion showed a significant difference ($p = 6.6e-09$), with VR players reporting a higher mean score (6.317) compared to non-VR players (5.317). The VR responses ranged narrowly from 5 (*Slightly Agree*) to 7 (*Strongly Agree*), with a median of 7, indicating consistently high levels of immersion. Non-VR responses spanned a wider range, from 4 (*Neutral*) to 7, with a median of 5, suggesting more variability in the immersion experience.

For *Progress Feedback*, VR players had a lower mean score (4.383) than non-VR players (5.033), with the difference being statistically significant ($p = 0.00467$). VR responses ranged from 3 (*Slightly Disagree*) to 6 (*Agree*), with outliers at both ends, indicating varied perceptions of progress indicators. In contrast, non-VR responses were more concentrated around 5, reflecting consistent agreement about the clarity of progress feedback in this setting.

Clarity of Goals also revealed a statistically significant difference ($p = 0.028$). VR players reported a mean score of 5.850, compared to 5.567 for non-VR players. The VR responses were tightly clustered around higher values, reflecting a clearer understanding of

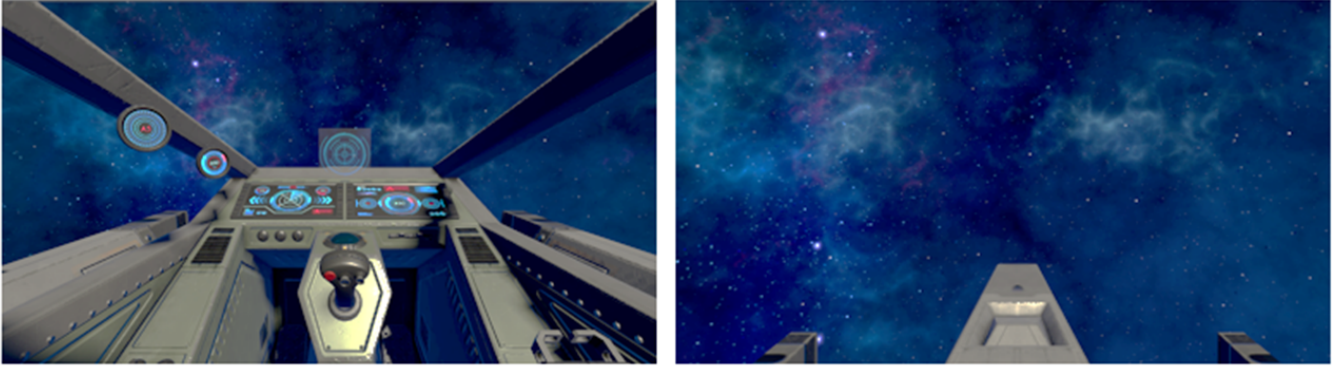


Figure 2: The default view from the VR player (left) and the non-VR player (right). The VR player sits inside the spaceship, experiencing from a first-person perspective, while the non-VR player observes from a slightly elevated viewpoint.

Table 1: Player Experience Criteria: Means, Differences, and Mann-Whitney U Test Results

Metric	Meaning	Curiosity	Mastery	Autonomy	Immersion
Mean (VR)	4.850	4.833	5.417	4.633	6.317
Mean (Non-VR)	4.683	4.783	5.300	4.517	5.317
Difference	0.167	0.050	0.117	0.117	1.000
U-statistic	1898.0	1626.5	1750.5	1709.0	2602.5
p-value	0.0933	0.993	0.452	0.621	6.6e-09

Metric	Progress Feedback	A-V Appeal	Challenge	Ease of Control	Clarity of Goals
Mean (VR)	4.383	5.133	4.950	5.317	5.850
Mean (Non-VR)	5.033	5.117	5.250	5.367	5.567
Difference	-0.650	0.017	-0.300	-0.050	0.283
U-statistic	1156.0	1685.5	1400.5	1685.0	1977.0
p-value	0.00467	0.719	0.172	0.717	0.028

game objectives, while non-VR responses displayed slightly more variation.

5.1.1 Player Platform Preference. After completing both tasks, participants indicated their platform preference. The majority (12 out of 20) expressed equal preference for VR and non-VR, suggesting that the shared-control mechanism successfully balanced the player experience across platforms. Four participants preferred VR, likely due to its immersive environment, while four favoured non-VR for its strategic gameplay. No participants expressed a strong preference for either platform, indicating minimal role imbalance.

5.2 Interview Findings

The interviews added a more nuanced understanding of the quantitative findings. Participants offered positive feedback to the collaborative nature of the game and the shared-control design, while also highlighting opportunities for improvement. Through thematic analysis, four key themes were identified, offering deeper insights into the player experience.

5.2.1 Gameplay Balance: Complementary Roles with Occasional Imbalances. Participants were generally positive about the balance between the VR and non-VR roles, emphasising the interdependence

and complementary nature of their tasks. The computer player’s role was described by participant A from Group 1 as “*equally important as the VR player, or even more important*”-G1_A, reflecting the strategic significance of the non-VR role. Another noted, “*the good thing is the two players’ roles are quite balanced*”-G2_A. Despite this positive feedback, some pointed out moments where the balance seemed uneven, such as when “*sometimes it feels like the VR player is doing more than the non-VR player*”-G6_B. This concern was further supported by the observation that “*the roles felt balanced, but it was difficult when one player was more experienced than the other*”-G7_A. These responses suggest that further refinements in task allocation and gameplay design could help ensure a more consistently equitable experience.

5.2.2 Immersion and Discomfort: The Strength and Weaknesses of VR Engagement. VR players described the experience as highly engaging, with many appreciating the novelty of being placed directly within the game world. The game was described as “*definitely more immersive for the VR player, this is inevitable due to the nature of VR*”-G1_A, underscoring the unparalleled sense of presence provided by the technology. Another highlighted the novelty, stating they “*prefer the VR role because that’s something I haven’t tried*

before"-G3_A, emphasising the unique appeal of VR games. However, challenges associated with VR immersion were also noted. Some participants mentioned that the intensity of VR gameplay could become overwhelming during fast-paced sequences, describing the experience as "great but it can get disorienting when there are too many fast-paced elements"-G4_B and "got tiring after a while"-G7_A.

A related issue was physical discomfort, particularly motion sickness, which was a recurring theme among VR players. Some described that they "didn't expect the level of dizziness during the game with VR"-G2_B. Fast movements and rapid turns were identified as significant contributors to this issue, with one noting that "faster turning made me feel a bit dizzy"-G4_B. These responses highlight the dual nature of VR immersion, as it enhances engagement but can also lead to sensory and physical strain, particularly in games with rapid motion.

5.2.3 Coordination and Communication: A Need for Improved Tools and Cues. Effective coordination was critical but challenging during shared-control gameplay. Participant G3_A described that they "enjoy the collaboration between players, but a bit difficult to communicate"-G3_A. Others suggested clearer cues, stating that "there should be clearer cues for when the VR player is moving the spaceship"-G4_A and that it was "sometimes hard to know when the VR player was moving, which affected my ability to aim as the non-VR player" -G9_B. Participants from Group 9 also proposed improvements to guidance and communication tools, with -G9_A suggesting that the "game could be improved with more guidance on how to communicate between players"-G9_A. These insights indicate that enhanced communication aids could improve player synergy.

5.2.4 Game Design Feedback: Suggestions for Enhancing Depth and Accessibility. Participants recommended refining the game to improve depth and accessibility. Suggestions included that "the game can be polished better; at the moment it's more like a demo, not an actual game"-G1_A, "maybe there can be more tasks assigned to each player"-G2_B, and "more missions or objectives could make it feel more complete"-G5_A. The need for tutorials was also emphasised, with one stating that "some tutorials for the game mechanics would help"-G7_A. Another proposed that "there's potential for more complex teamwork mechanics in the game, like joint puzzles or shared decision-making"-G9_B. These suggestions highlight opportunities to enhance player engagement and collaboration.

6 Discussion

This study investigated the potential for shared-control mechanisms as a novel solution for balancing VR and non-VR player experience in cross-platform multiplayer gameplay. The shared-control design effectively facilitated collaboration and interdependence between players which appeared to impact players' sense that despite their play modality, their role was essential for success. Quantitative PXI evaluations showed consistent scores for aspects like *Mastery*, *Autonomy*, and *Ease of Control* across VR and non-VR participants. Participants expressed equal preferences for both platforms, indicating that the roles were engaging and balanced—though these conclusions are preliminary given the study's design. However, some platform-specific disparities persisted, particularly in *Immersion*, *Progress Feedback*, and *Clarity of Goals*. VR players reported

higher immersion scores, reflecting the inherent capabilities of VR technology to provide engrossing experiences. The task-oriented nature of the non-VR role provided clearer and more immediate feedback, while VR players found it harder to assess their success. Non-VR participants on the other hand, reported a segmented understanding of their tasks, often lacking awareness of how their role fits into the broader game objectives.

6.1 Addressing the Research Questions

RQ1: To what extent can a shared-control mechanism enable balanced engagement and equitable participation between VR and non-VR players in cross-platform games? The shared-control mechanism successfully promoted balanced engagement and equitable participation. Quantitative PXI results for *Mastery*, *Autonomy*, and *Ease of Control* showed no significant differences between VR and non-VR players, while participant preferences reflected similar satisfaction across roles. However, persistent disparities in *Immersion* and *Progress Feedback* highlight the inherent challenges of bridging sensory and interaction differences across platforms. Despite these gaps, the shared-control mechanism proved effective in creating an inclusive and collaborative experience.

RQ2: How do players perceive their roles and interactions when engaging in a shared-control mechanism across different platforms? Participants viewed their roles as distinct but equally important. VR players emphasised the immersive and intuitive aspects of piloting, while non-VR players appreciated the strategic and task-oriented nature of their role. Both groups highlighted the importance of collaboration, noting that success depended on effective communication and coordination. These findings confirm that shared control fosters teamwork and balances contributions, addressing traditional disparities in cross-platform designs.

6.2 Broader Implications for Cross-Platform Game Design

The shared-control mechanism offers a novel approach to address engagement disparities in cross-platform games. By distributing control over a single game entity, the model promotes mutual dependence and collaboration, ensuring both roles are equally critical. This stands in contrast to traditional asymmetrical designs, which often prioritise VR players and leave non-VR participants in passive or secondary roles.

Beyond gaming, shared-control mechanisms have broader applications in areas like education, remote collaboration, and hybrid digital workspaces. They demonstrate how balanced participation can be achieved through thoughtful design, creating environments where all participants actively contribute. Previous approaches, such as secondary displays or physical interactions [6], often failed to engage non-VR participants fully. By contrast, the shared-control model integrates both players into the core gameplay, ensuring that each has a meaningful and active role.

6.3 Limitations and Future Directions

Despite its successes, the shared-control mechanism has limitations. The sensory gap in *Immersion* between VR and non-VR players remains a significant challenge, tied to the inherent strengths of

VR technology. Enhancing non-VR engagement through richer content, visual feedback, or role-switching mechanics could help mitigate this disparity. The segmented perception of tasks for non-VR participants also highlights the need for clearer integration of roles to improve their connection to the overall game objectives.

Another key limitation is the potential influence of gameplay design on *Progress Feedback* ratings. The non-VR player's task naturally provides more immediate confirmation of actions, which may contribute to higher ratings in certain aspects of the experience. Addressing this issue would require further investigation, such as refining the study's design to isolate the impact of platform differences or modifying gameplay mechanics to offer comparable levels of feedback across roles.

Future research could explore adaptive designs that dynamically adjust tasks and feedback systems based on player roles, further balancing the experience. Additionally, expanding shared-control mechanisms to different game genres could reveal new possibilities for cooperative and inclusive gameplay.

7 Conclusions

This research introduced a shared-control mechanism to address engagement imbalances in cross-platform games, enabling collaboration between VR and non-VR players. As implemented in our game, the mechanism required players to collectively control a single entity, fostering interdependence and balanced gameplay. A user study (N=20) showed that the mechanism successfully created engaging experiences for both platforms. VR players highlighted the immersive nature of piloting, while non-VR players valued the strategic and immediate feedback of shooting. Despite these successes, some disparities remained. VR players reported higher *Immersion*, while non-VR participants rated *Progress Feedback* more favourably. These differences reflect inherent platform strengths but underscore the need for further refinement in integrating sensory and feedback mechanisms. This work demonstrates the potential of shared-control gameplay to foster balanced engagement and mutual dependence in cross-platform games. By addressing these challenges, shared-control mechanisms provide a foundation for more inclusive and dynamic gameplay experiences, paving the way for broader adoption of cross-platform and VR gaming.

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